Air University, Islamabad

Software Requirement Specification
(SRS DOCUMENT)

for

EZSell – a Marketplace for Buying & Selling Pre-Owned Items

Version 1.0

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Revision History

Name	Date	Reason for Changes	Version

Application Evaluation History

Comments (by committee) ***include the ones given at scope time both in doc and presentation**	**Action Taken**

Supervised by

Supervisor's Name

Signature_______________

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1. Introduction

This Software Requirements Specification (SRS) outlines the functional and non-functional requirements of **EZSell**, an online platform that facilitates the buying and selling of secondhand items, particularly electronic gadgets and furniture. The document is designed to serve as a comprehensive guide for developers, testers, project managers, UI/UX designers, system architects, and other stakeholders involved in the development and maintenance of EZSell.

1.1 Purpose

The purpose of this SRS is to formally define the requirements of the EZSell platform. It is intended for a variety of audiences including:

- **Developers**: to understand what needs to be implemented.
- **Project Managers**: to plan and monitor the development.
- **UI/UX Designers**: to align interface elements with functional needs.
- **Testers**: to derive test cases from defined functionalities.
- **Documentation Writers**: to prepare user manuals and help content.
- **Marketing Teams**: to align product features with promotion strategies.
- **End Users**: indirectly, as it defines what features they will interact with.

1.2 Scope

EZSell is a web-based application that allows users to **post, search, and manage advertisements** for secondhand items such as **mobiles, tablets, laptops, and furniture**. It incorporates **AI-based price prediction**, **AR-based customization for furniture placement**, **secure user registration**, **ad fraud prevention** and monitors user's engagement. The system is designed to fill key gaps in existing marketplaces by offering better pricing accuracy, fraud prevention, and user experience.

Key features include:

- AI-driven dynamic pricing engine
- AR try-on for used furniture
- User-friendly ad posting and management
- Smart filters and product recommendations
- Fraud detection mechanisms for secure listings
- Administrative moderation tools

1.3 Modules

The major modules of the EZSell platform include:

- **M1: Profile Management Module** – Handles user authentication, profile creation, and verification.
- **M2: Product Listings and Management Module** – Allows posting, editing, and deletion of ads.
- **M3: Price Prediction Module** – Predicts fair prices using AI based on item features.
- **M4: Recommendations Module** – Suggests relevant listings using filters and AI.
- **M5: Analytical Dashboard Module** – Displays insights into user activity and trends.
- **M6: Customization Try-On Module** – Lets users overlay furniture images on room layouts.
- **M7: Fraud Prevention Module** – Detects and flags suspicious listings and users.

Each module is uniquely labeled (e.g., M1, M2) for traceability throughout the system lifecycle.

1.4 Overview

The remainder of this document is organized as follows:

2. Overall Description – Product perspective, user classes, operating environment, and constraints. **3. Requirement Identifying Technique** – Techniques used to gather software requirements. **4. Functional Requirements** – Detailed list of core functions the system must perform. **5. Non-Functional Requirements** – Performance, usability, security, and reliability. **6. External Interface Requirements** – User, software, hardware, and communication interfaces. **7. References** – Supporting books, articles, and web resources.

2. Overall Description

This section presents a high-level overview of the product and the environment in which it will be used, the anticipated users, and known constraints, assumptions, and dependencies.

2.1 Product Perspective

EZSell is an entirely new platform developed as a web-based application for facilitating the buying and selling of second-hand items. It is not a continuation of an existing product but has been built from scratch with advanced features such as AI-based pricing and AR-based furniture try-on. The platform connects buyers and sellers via secure and streamlined interactions. Visual models, such as context diagrams and system DFDs, are included in Appendix A to show system interrelations.

2.2 User classes and characteristics

EZSell is designed to accommodate a variety of user classes:

- **General Users (Buyers and Sellers):** Individuals interested in purchasing or selling used electronics and furniture. They may not have technical expertise but require a smooth and intuitive interface.
- **Administrators:** Staff responsible for managing listings, verifying users, and handling flagged content. They need full access to moderation tools.
- **System Developers/Testers:** Personnel working on the development and maintenance of EZSell, requiring technical documentation and access to API

2.3 Operating Environment

- OE-1: The system shall operate correctly on modern web browsers including:
    - Google Chrome (v85+)
    - Mozilla Firefox (v78+)
    - Microsoft Edge (v85+)
    - Safari (v13+)
- OE-2: Databases will be hosted using PostgreSQL or MySQL, integrated via cloud platforms like AWS or Heroku.
- OE-3: The application shall be accessible via desktop and mobile browsers across all major devices.

2.4 Design and Implementation Constraints

- CON-1: The backend must be developed using Django as the primary web framework.
- CON-2: The frontend shall be built using React.js for component-based design and responsiveness.
- CON-3: The application must implement user verification via email or phone to prevent fraudulent activity.
- CON-4: AI-based price predictions shall be implemented using Python and scikit-learn.
- CON-5: The system must be hosted on secure cloud platforms like Heroku or AWS for deployment and scaling.
- CON-6: The customization feature must support only basic furniture positioning (move/rotate), with no object resizing or color changing to maintain realism of used items.
- CON-7: All textual and visual content shall be stored securely using SQL and cloud storage options.

3. Requirement Identifying Technique

• Context Diagram

The EZSell Marketplace System is a new online platform that replaces traditional and informal methods of buying and selling second-hand items, such as in-person haggling or using unmoderated social media groups. The platform provides a secure, AI-enhanced, and user-friendly environment for posting, discovering, and purchasing pre-owned products.

The **context diagram** in **Figure A-1** illustrates the key external entities that interact with the EZSell system, such as buyers, sellers, administrators, and third-party services. These entities communicate with the EZSell Marketplace System through various interfaces — including product listings, search requests, image uploads, fraud detection alerts, AR visualization, and messaging services — to support end-to-end marketplace functionality.

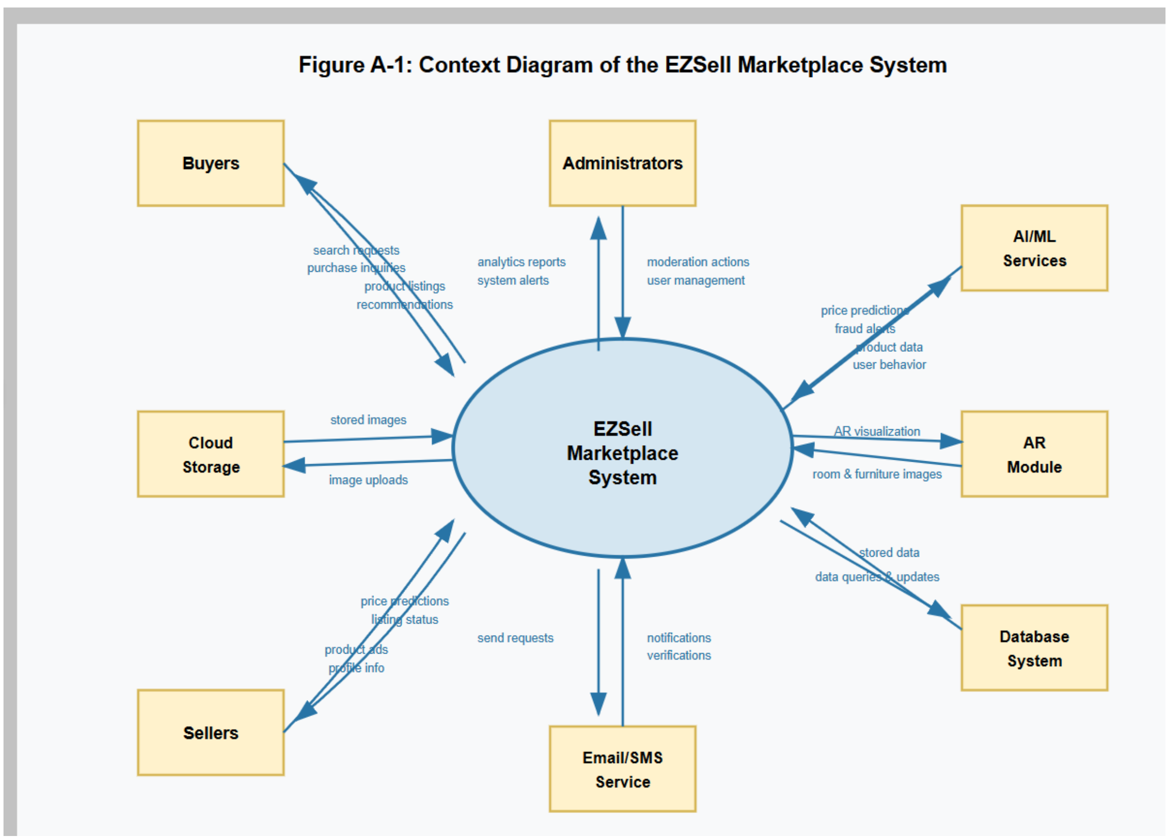

Figure A-1: Context diagram of the Cafeteria Ordering System.

- ## User Classes and Characteristics

Table A-1 Shows user classes and characteristic for EZSell

User class	Description
Buyer	A **Buyer** is a registered user looking to purchase second-hand products such as electronics, furniture, or other listed items. Buyers may browse listings, use search filters, contact sellers, and finalize purchases via chat or "Buy Now" functionality. An estimated 60% of buyers will access the platform through mobile devices, while others may use desktop or tablets.
Seller	A **Seller** is a verified user who lists used items for sale. Sellers must provide item details, upload product images, and may use the AI-powered price suggestion feature. Sellers are notified upon buyer interest and can negotiate deals via the in-app chat. Some sellers may post multiple items per week and require guidance on image formats, pricing, and fraud alerts.
Admin	The **Admin** oversees platform moderation, reviews flagged listings, handles reports, and manages fraudulent or suspicious activity. Admins have access to user behavior logs and fraud detection dashboards. They manually review posts when AI raises suspicion and have authority to approve, reject, or request changes to listings.
AI Pricing Engine	The **AI Pricing Engine** is an internal system component, not a human user, that provides automated price suggestions based on product specifications and current market data. It interacts with listings before publishing, ensuring price consistency. If unavailable, sellers can input prices manually.
AR Visualization Module	The **AR Visualization Module** allows buyers to preview how products—especially furniture or room decor—would appear in their actual environment using Augmented Reality. The module receives room layout and object images and renders a virtual placement. Mostly accessed via mobile devices and requires camera permissions.

- **Use case Diagram**

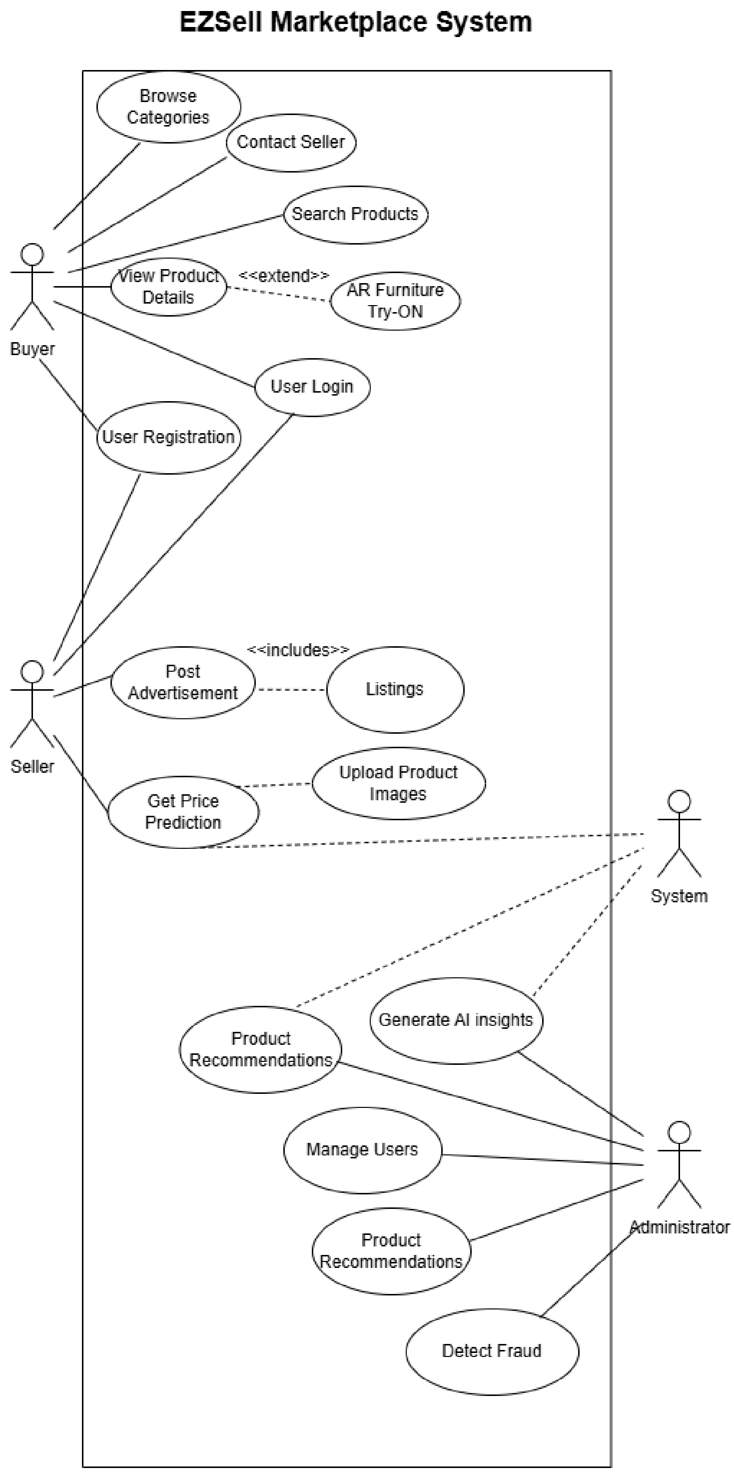

Figure A-2: Use Case Diagram of EZSell

- # Detaiied Use Cases:

Table A-2: Detailed Use Case Diagram of EZSell

Use Case 1: Post a Product for Sale

Field	Description
Use Case ID	UC-01
Use Case Name	Post a Product for Sale
Primary Actor	Registered Seller
Goal	Allow users to list their pre-owned product with details, images, and AI-generated price suggestions.
Preconditions	• Users must be registered and logged in. • Seller account must be verified.
Postconditions	- Product is listed in the marketplace. - Listing is subject to fraud detection before becoming visible.
Main Flow	1. Seller clicks "Post New Product". 2. System shows a form with fields: product title, description, category, price, images, etc. 3. Seller fills in the form and uploads images. 4. System uses AI to suggest a price based on product data. 5. Seller accepts or edits suggested price. 6. System runs the fraud detection module. 7. If passed, product is posted and visible to buyers. 8. Confirmation message is displayed.
Alternate Flows	4a. If AI fails to suggest a price → default to manual entry. 6a. If flagged by fraud detection → notify admin and hold post for moderation.
Exceptions	- Image format unsupported → show error. - Incomplete fields → prompt user to complete the form.
Priority	High

Use Case 2: Search and Purchase a Product

Field	**Description**
Use Case ID	UC-02
Use Case Name	Search and Purchase a Product
Primary Actor	Registered Buyer
Goal	Enable buyers to search for products, view details, and initiate a purchase.
Preconditions	• User must be registered and logged in. • Internet connection is active.
Postconditions	• Product is marked as “in negotiation” or “sold”. • Seller is notified of buyer interest.
Main Flow	1. Buyer navigates to the homepage or search bar. 2. Enters keywords or selects filters (category, location, price range). 3. System displays search results with thumbnails and summaries. 4. Buyer clicks on a product to view full details. 5. Buyer clicks “Contact Seller” or “Buy Now” depending on listing type. 6. System facilitates chat or forwards inquiry to seller. 7. Seller confirms availability and negotiates if needed. 8. Buyer confirms purchase. 9. Product marked as sold; system sends confirmation emails to both parties.
Alternate Flows	5a. AR try-on available for furniture → buyer launches AR preview before purchase. 6a. Seller doesn’t respond in 24 hours → automated follow-up sent.
Exceptions	- No products found → show alternative recommendations. - Chat fails → display error and retry option.
Priority	High

Use Case 3: Detect Fraudulent Listing

Field	Description
Use Case ID	UC-03
Use Case Name	Detect Fraudulent Listing
Primary Actor	System (Fraud Detection Module), Admin
Goal	Automatically detect and prevent fraudulent listings before they go live.
Preconditions	• Product has been submitted for listing. • Fraud detection service is active.
Postconditions	• Listing is either approved, flagged, or rejected. • Admin is notified for manual review if necessary.
Main Flow	1. Seller submits product listing. 2. System sends listing data to fraud detection module. 3. Module analyzes text (e.g., suspicious keywords), image metadata, seller behavior history. 4. If score is within safe threshold → approve post. 5. If score is suspicious → flag post for admin. 6. Admin receives flagged post in dashboard and manually reviews. 7. Admin approves, edits, or rejects the listing.
Alternate Flows	3a. If model confidence is low → send directly to admin without automatic decision.
Exceptions	- Fraud model service unavailable → fallback to manual review.
Priority	Medium-High

- ## Event-Response Tables

The EZSell system has to deal with various events, including these:

1. **User navigates to homepage or product listings.**
→ System detects activity and loads recent or recommended ads.
→ Event triggers listing display and analytics logging.

2. **User initiates "Post an Ad" action (button click or shortcut).**
→ System transitions to ad creation form, initializes category fields.
→ Data like session, user ID, and previous drafts may be loaded.

3. **User clicks "Get Price Suggestion" for AI price prediction.**
→ System processes product details and fetches pricing.
→ Response includes confidence score and suggestions.

4. **Ad expiration timer completes (e.g., 30 days).**
→ System auto-expires the ad, notifies the seller, and optionally archives it.
→ Could also relate to session timeout or fraud review deadline.

Table A-3 Event-Response Table for EZSell

Table 1: User Interface and Navigation Events

Event	System State	Response	Frequency
User clicks "Post Ad" button or uses Alt+P shortcut	User is logged in and authenticated. Main dashboard is displayed.	1. Verify user authentication status 2. Display "Create New Advertisement" form 3. Show category selection options 4. Initialize form validation system Data Elements: User ID, Session Token Resulting State: Advertisement form displayed, awaiting category selection	High (Multiple per session)
User selects product category (Electronics/Furniture)	Advertisement form is displayed with category options available.	1. Load category-specific specification fields 2. Display relevant input forms (Electronics: Brand, Model, Storage; Furniture: Material, Dimensions) 3. Enable real-time validation for selected category Data Elements: Category Type, Field Templates Resulting State: Category-specific form displayed, ready for specification entry	Medium (Once per ad)
User enters product specifications and description	Category-specific form is displayed with empty specification fields.	1. Validate input format in real-time 2. Provide dropdown suggestions where applicable 3. Show tooltips and guidance messages 4. Prepare data for AI price prediction Data Elements: Product Title, Description, Brand, Model, Condition, Specifications Resulting State: Specifications	Medium (Once per ad)

Event	System State	Response	Frequency
		entered, form ready for image upload	

Table 2: Image Upload and Media Management Events

Event	System State	Response	Frequency
User uploads product images	Product specifications are entered. Image upload section is available.	1. Validate image format (JPG, PNG, WebP) and size (<5MB) 2. Upload to cloud storage (AWS S3/Cloudinary) 3. Generate thumbnails for display 4. Run basic image analysis for fraud detection 5. Allow image reordering Data Elements: Image Files, File Metadata, Upload Progress Resulting State: Images uploaded and validated, ready for AI price prediction	Medium (1-5 per ad)
Image upload fails	User attempting to upload image. Network timeout, unsupported format, or file size exceeds limit.	1. Display specific error message (format, size, or network issue) 2. Provide options to retry, modify image, or continue 3. If minimum 1 image uploaded, allow continuation 4. Save form draft if user cancels Data Elements: Error Type, File Information, Draft Data Resulting State: Error displayed with recovery options, or draft saved	Low (5-10% of uploads)

Table 3: AI Price Prediction and Pricing Events

Event	System State	Response	Frequency
User clicks "Get Price Suggestion"	Product specifications and images are uploaded. AI Price Prediction System is operational.	1. Send specifications to AI Price Prediction Module (M3) 2. Display loading indicator 3. Wait for AI response (max 1.5 seconds per PER-2) 4. Prepare for price display Data Elements: Product Specs, Market Data Request Resulting State: AI prediction request in progress	High (Once per ad)
AI Price Prediction System responds with suggestion	AI prediction request is in progress. System awaiting response within 1.5 seconds.	1. Display suggested price range with confidence level 2. Show market analysis and justification 3. Provide options to accept or set custom price 4. Log prediction data for system learning Data Elements: Price Range, Confidence Level, Market Analysis Resulting State: Price suggestion displayed, awaiting user decision	High (85% success rate)
User accepts AI price suggestion	AI price suggestion is displayed with confidence level and market analysis.	1. Set suggested price as final asking price 2. Validate price is within market range 3. Continue to contact information section 4. Update price acceptance statistics Data Elements: Accepted Price, User Choice Statistics Resulting State: Price confirmed, ready for contact details	High (70% acceptance)
User rejects AI suggestion and sets custom price	AI price suggestion is displayed. User chooses custom pricing	1. Display custom price input field 2. Validate against market extremes (10%-300% of AI suggestion)	Medium (30% custom

Event	System State	Response	Frequency
	option.	3. Show warning if price significantly deviates 4. Allow override with additional verification if needed Data Elements: Custom Price, Validation Range, Override Request Resulting State: Custom price set, ready for contact details	pricing)
AI Price Prediction System unavailable	User requests price suggestion. AI system timeout or error occurs.	1. Display system unavailable message 2. Offer retry after 30 seconds 3. Provide alternative: market price range from recent listings 4. Allow user to continue without AI suggestion or save draft Data Elements: Error Code, Fallback Price Data, User Choice Resulting State: Fallback pricing provided or process saved as draft	Low (15% system downtime)

Table 4: Security and Fraud Detection Events

Event	System State	Response	Frequency
Fraud Detection System identifies suspicious patterns	User submits final advertisement. System runs fraud detection analysis.	1. Hold advertisement from going live 2. Display message: "Your listing is under review for quality assurance" 3. Flag for administrator review within 24 hours 4. Generate alert with suspicious pattern details Data Elements: Fraud Score, Pattern Details, Admin Alert Resulting State: Advertisement held in review queue	Low (5% of ads)
Administrator approves flagged advertisement	Advertisement is in review queue. Administrator completes manual review.	1. Change advertisement status to "Active" 2. Make advertisement live and searchable 3. Send approval notification to seller 4. Update fraud detection learning data Data Elements: Admin Decision, Approval Reason, Notification Data Resulting State: Advertisement live and active	Low (60% of flagged)
Administrator rejects flagged advertisement	Advertisement is in review queue. Administrator identifies policy violation.	1. Keep advertisement status as "Rejected" 2. Send detailed rejection explanation to seller 3. Provide opportunity to revise and resubmit 4. Update user violation history Data Elements: Rejection Reason, Revision Guidelines, User History Resulting State: Advertisement rejected, seller notified with revision option	Low (40% of flagged)

Table 5: System Events and Error Handling

Event	System State	Response	Frequency
User session expires during advertisement posting	User is in middle of posting process. Session timeout occurs.	1. Save current form data as draft 2. Redirect to login page with recovery message 3. Upon re-authentication, offer to restore saved data 4. Continue from last completed step if restored Data Elements: Draft Data, Session Info, Recovery Token Resulting State: User redirected to login with recovery option	Medium (10-15% of sessions)
System validates invalid input data	User submits form with invalid data (price out of range, missing required fields).	1. Highlight invalid fields with specific error messages 2. Prevent form submission until corrected 3. Provide guidance for acceptable values 4. Maintain other valid data in form Data Elements: Validation Errors, Field Values, Guidance Text Resulting State: Form displayed with validation errors, awaiting correction	Medium (20-30% of submissions)
Advertisement posting completes successfully	All validations pass. Advertisement ready to go live.	1. Generate unique advertisement ID 2. Store advertisement in database with "Active" status 3. Make advertisement searchable by buyers 4. Send confirmation email/SMS to seller 5. Schedule automatic expiration based on duration 6. Display success confirmation with ID Data Elements: Ad ID, Confirmation Details, Expiration Schedule Resulting State: Advertisement live and active, seller notified	High (85-90% success)

4. Functional Requirements

This section describes the functional requirements of the EZSell platform organized by major system features. Each requirement is written from a system or user perspective and includes its source, rationale, and any dependencies.

4.1 Functional Requirements per module

4.1.1 Functional Requirement: FR-1 - User Registration and Login

Identifier	FR-1
Title	User Registration and Login
Requirement	The user shall be able to register and log in using email or phone number with a verification mechanism to securely access their account.
Source	Client Requirement Document
Rationale	To authenticate users securely and ensure unique user identities.
Business Rule	Email or phone number verification is mandatory during registration.
Dependencies	None

4.1.2 Functional Requirement: FR-2 - Product Listing Creation

Identifier	FR-2
Title	Product Listing Creation
Requirement	The seller shall be able to post product ads with title, category, description, and up to 5 images for each listing.
Source	Stakeholder Interviews
Rationale	Allows users to sell used items on the platform effectively.
Business Rule	Each listing can have a maximum of 5 images.
Dependencies	FR-1

4.1.3 Functional Requirement: FR-3 - AI Price Prediction

Identifier	FR-3
Title	AI Price Prediction
Requirement	When product specifications are submitted, the system shall provide an AI-generated price recommendation based on brand, year, and condition.
Source	Project Scope Document
Rationale	To assist sellers in fair and competitive pricing using real-time data trends.
Business Rule	Prediction must consider market price trends and similar listings.
Dependencies	FR-2

4.1.4 Functional Requirement: FR-4 - Smart Product Search and Filter

Identifier	FR-4
Title	Smart Product Search and Filter
Requirement	The user shall be able to search and filter products by category, price range, brand, and location.
Source	Competitor Platform Analysis
Rationale	To enhance discoverability of listings and improve user experience.
Business Rule	The filter must allow multi-selection in each category.
Dependencies	FR-2

4.1.5 Functional Requirement: FR-5 - Marketplace Analytics Dashboard

Identifier	FR-5
Title	Marketplace Analytics Dashboard
Requirement	The admin shall be able to view marketplace analytics such as top-performing listings, engagement rates, and category popularity through visual reports.
Source	Admin Requirements
Rationale	To monitor platform trends and improve decision-making.
Business Rule	Admins must be authenticated to access the dashboard.
Dependencies	FR-2, FR-6

4.1.6 Functional Requirement: FR-6 - Engagement Logging

Identifier	FR-6
Title	Engagement Logging
Requirement	The system shall track user browsing behavior and interaction with listings to generate personalized engagement insights.
Source	Data Analysis Objective
Rationale	To understand user preferences and personalize recommendations.
Business Rule	Tracking must comply with privacy and cookie policies.
Dependencies	FR-4

4.1.7 Functional Requirement: FR-7 - AR Furniture Visualization

Identifier	FR-7
Title	AR Furniture Visualization
Requirement	The user shall be able to upload a room image and overlay a selected furniture image with controls to reposition and rotate it within the view.
Source	Client Feedback
Rationale	To help users visualize used furniture in their own spaces.
Business Rule	Feature supports only furniture category listings.
Dependencies	FR-2

4.1.8 Functional Requirement: FR-8 - Suspicious Activity Detection

Identifier	FR-8
Title	Suspicious Activity Detection
Requirement	The system shall automatically analyze listings for anomalies and flag those that may be fraudulent based on patterns.
Source	Platform Security Policy
Rationale	To prevent scams and platform credebillity.
Business Rule	Listings must be reviewed by admin if flagged.
Dependencies	FR-1, FR-2

5. Non-Functional Requirements

This section outlines the quality attributes of the EZSell system, including reliability, usability, performance, and security.

5.1 Reliability

- **REL-1**: The system shall have a Mean Time Between Failures (MTBF) of at least 500 hours during normal operation.
- **REL-2**: The system shall automatically log any critical failure and notify the admin within 10 minutes.
- **REL-3**: In case of a failure, the system shall allow restoration of user and listing data within 1 hour using database backups.

5.2 Usability

- **USE-1**: The system shall allow new users to complete registration and post their first ad within 3 minutes without prior training.
- **USE-2**: The interface shall provide tooltips and field validation to minimize user input errors during product listing.
- **USE-3**: The platform shall be accessible on desktops using responsive design to ensure ease of use.

5.3 Performance

- **PER-1**: 95% of all webpages shall fully load within 3 seconds over a 10 Mbps or faster connection.
- **PER-2**: Price prediction results shall be displayed within 1.5 seconds after input submission.
- **PER-3**: The search results with applied filters shall be generated within 2 seconds for 90% of user queries.

5.4 Security

- **SEC-1**: The system shall verify user identity via email or phone verification before allowing ad submission.
- **SEC-2**: All sensitive user data (passwords, contact details) shall be encrypted.
- **SEC-3**: The platform shall automatically flag and temporarily restrict ads suspected of fraudulent content until reviewed by an admin.
- **SEC-4**: Admin access to system controls shall be protected via two-factor authentication.

6. External Interface Requirements

This section outlines the interfaces between the EZSell platform and external components, including user-facing interfaces, supporting software systems, hardware environments, and communication protocols. These interfaces ensure smooth interaction between users, subsystems, and external services while maintaining performance, usability, and security. Each subsection defines specific requirements for integration, consistency, and accessibility across the entire platform.

6.1 User Interfaces Requirements

- **UI-1**: The interface shall follow a clean, minimalistic layout with consistent font styles, colors, and control icons across all pages.
- **UI-2**: All pages shall support responsive design and adapt to desktop, tablet, and mobile screen resolutions (minimum 360x640).
- **UI-3**: Navigation bar with links to Home, Categories, Post Ad, Search, and Profile shall appear on every screen.
- **UI-4**: System shall use tooltips, placeholder text, and validation messages to guide the user input process.
- **UI-5**: Accessible design principles will be applied for visually impaired users, including font scaling and contrast settings.
- **UI-6**: Shortcut keys will be provided for major actions like "Post Ad" (Alt + P), "Search" (Alt + S), and "Login" (Alt + L).
- **UI-7**: Message and error notifications shall be displayed as modal alerts or inline banners.

6.2 Software interfaces

- **SI-1**: The system shall integrate with a e.h **PostgreSQL or MySQL** database to store and retrieve user profiles, product listings, and transactions.
- **SI-2**: The system shall communicate with an **AI-based price prediction module** using RESTful API (v1.0).
- **SI-3**: The platform shall use e,g **Django REST Framework** for backend API creation and exposure.
- **SI-4**: Frontend shall be built in **React.js** and connected to backend endpoints for data operations.
- **SI-5**: Integration with e.g **AWS S3 or Cloudinary** for image upload and storage services.

6.3 Hardware interfaces

- **HI-1**: The system is a web application and shall run on any standard hardware with a modern web browser.
- **HI-2**: Servers hosting the backend (e.g., AWS EC2) shall support deployment of e,g Django with database and AI model.
- **HI-3**: Frontend users require no special hardware, only a device capable of rendering HTML5 content.

6.4 Communications interfaces

- **CI-1**: The system shall send email and SMS notifications to users for verification, password recovery, and listing updates.
- **CI-2**: Communication between frontend and backend shall occur over secure **HTTPS** using **APIs**.
- **CI-3**: WebSocket protocol may be used in future to enable real-time chat between buyer and seller.

7. References

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        - ***Introduction → Purpose & Scope* (general software engineering practices)**
        - ***Design and Implementation Constraints* (standard development methods)**
        - ***Functional & Non-Functional Requirements* structure**
    - **Link: Foundation for SRS writing methodology and software lifecycle processes.**
2. **Stuart Russell & Peter Norvig – *Artificial Intelligence: A Modern Approach***
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    - **Link: Supports justification and theory behind using AI/ML in pricing and behavior detection.**
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        - ***Price Prediction Algorithms***

    - *Search and Filtering Features*
  - **Link: Underpins algorithmic processes used in the backend of smart filtering and AI-based recommendations.**

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   - **Used in:**
     - *AI Pricing and Recommendations Module*
   - **Link: Technical basis for predictive modeling, especially if time-based trends are used.**
2. **LeCun, Bengio, Hinton –** ***Deep Learning***
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     - *Furniture Customization Module (image overlay, AR features)*
   - **Link: Justifies the use of image classification or embedding features in customization.**

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    - ***User Interface Requirements***
    - ***Non-Functional Requirements: Usability***
- **Link: Justifies design standards and importance of user experience.**

2. **M. Richardson – *Beautiful Soup Documentation (https://www.crummy.com/software/BeautifulSoup/bs4/doc/, [Accessed: Feb 5, 2025].)***
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        - ***Web Scraping Techniques for Data Collection***
    - **Link: Technical reference for scraping used product data from marketplaces.**
3. **Brainly – *Shortcomings of OLX (https://brainly.in/question/53661651, [Accessed: Mar. 15, 2025)***
    - **Used in:**
        - ***Introduction, Market Analysis, Justification of Product***
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